

La Vogue Vista: a cross-platform AI-powered mobile platform for real-time virtual beauty try-on and personalized recommendation

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Abstract. La Vogue Vista is a cross-platform mobile system designed to improve digital beauty product evaluation through real-time augmented reality (AR) visualization and AI-assisted recommendation. The application integrates Flutter for multi-platform user experience, Firebase services for authentication and catalog persistence, and native iOS/Android rendering pipelines for low-latency camera processing. MediaPipe landmark tracking and TensorFlow Lite segmentation are used to align cosmetic overlays and hair recoloring masks to live facial regions while preserving texture and lighting continuity. A cloud-assisted recommendation module, supported by OpenRouter-based language reasoning over structured visual descriptors, provides explainable style and product suggestions. The architecture follows a hybrid strategy: latency-critical geometric tracking and rendering are executed on-device, while higher-level semantic recommendation is processed through cloud inference. Experimental implementation on physical devices demonstrates interactive live operation for lipstick and hair try-on workflows and practical support for recommendation-guided product exploration. The proposed system shows that combining edge computer vision, native AR rendering, and AI recommendation in a single mobile platform is technically feasible and commercially relevant for personalized beauty e-commerce.

Keywords: Artificial Intelligence, Augmented Reality, Computer Vision, Edge AI, Mobile Commerce

1 Introduction

E-commerce beauty selection is difficult because 2D product swatches do not accurately represent real appearance across different skin tones, facial geometry, and lighting conditions. This mismatch increases user uncertainty and weakens purchase confidence. Existing solutions are often fragmented, where one application focuses only on makeup, while another separately addresses hair simulation or recommendation. This project addresses that gap by proposing a single mobile platform that integrates real-time AR try-on and AI-supported personalization.

La Vogue Vista is designed to provide a complete decision-support workflow: user onboarding, product browsing, live visual simulation, and recommendation-guided product selection. The project emphasizes practical usability, mobile responsiveness, and modular architecture suitable for incremental feature extension.

2 Literature review

Recent AR beauty systems commonly rely on facial landmarks and semantic segmentation. Landmark-based methods offer efficient region anchoring for overlays, while segmentation supports pixel-level recoloring and masking. MediaPipe has become widely adopted in mobile vision due to speed and deployment simplicity, and TensorFlow Lite enables low-footprint on-device inference suitable for interactive use.

At the same time, recommendation systems in this domain increasingly incorporate language-model reasoning for explainable guidance. However, many implementations separate visual simulation from recommendation generation, resulting in inconsistent user experience. This project adopts a hybrid approach that combines real-time vision pipelines with semantic recommendation support in one application flow.

3 Methodology

3.1 System architecture

The system uses Flutter UI and workflow orchestration, Firebase Auth/Firestore/S storage for user and product data, native camera/rendering layers (Swift/Kotlin), MediaPipe and TFLite for on-device computer vision, and OpenRouter LLM reasoning for explainable recommendations.

3.2 Geometric AR formulation (makeup and hair)

Given normalized landmarks $p_i=(x_i,y_i)$, region alignment is computed by:

$$c=(1/N) \sum_{i=1..N} p_i, \\ \theta=\text{atan2}(y_b-y_a, x_b-x_a), \\ s=\|p_b-p_a\| / d_0,$$

where c is anchor center, θ is orientation, and s is scale relative to template distance d_0 .

Overlay transform:

$$T=\text{Translate}(c) * \text{Rotate}(\theta) * \text{Scale}(s)$$

Hair mask compositing uses confidence $m(u,v)$ in $[0,1]$:

$$I_{\text{out}}=(1-\alpha*m)I_{\text{in}} + \alpha*m*I_{\text{shade}}$$

with texture-preserving luminance control in HSL space:

$$L_{\text{out}}=\max(L_{\text{in}}, \lambda*L_{\text{target}}), \quad 0<\lambda<1.$$

3.3 Recommendation inference

Structured visual descriptors (face geometry ratios, tone features, hair attributes) are converted into prompt-safe metadata and passed to a cloud LLM service for explainable recommendation generation.

4 Results and discussion

The developed system supports integrated shopping and try-on workflows and achieves interactive behavior on physical mobile devices for lipstick and hair try-on use cases. User interaction remains responsive during live preview under typical indoor lighting and moderate pose changes. The hybrid architecture reduces visual latency by retaining geometric tracking and rendering at the edge, while still enabling richer recommendation narratives through cloud reasoning.

Observed engineering trade-offs include sensitivity to camera quality, illumination variation, and segmentation confidence under rapid motion. Stability benefits from balancing inference frequency, smoothing strength, and rendering update cadence. The modular design allows progressive enhancement of individual features without replacing the complete pipeline.

5 Conclusion and recommendation

La Vogue Vista demonstrates that a unified mobile beauty platform combining AR visualization and AI recommendation is technically feasible and practically useful for digital product evaluation. The project contributes a deployable architecture that links edge vision processing with cloud-assisted semantic reasoning in a single user experience.

Future work should include broader device benchmarking, controlled user studies on recommendation usefulness, and continued robustness improvements for challenging real-world conditions. The approach can also be extended to additional cosmetic categories and personalized recommendation refinement.

References

- [1] Google AI Edge, "MediaPipe Solutions: Face Landmarker," Google, 2024. [Online]. Available: https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker. Accessed: Apr. 30, 2026.
- [2] Google AI Edge, "MediaPipe Solutions: Hand Landmarker," Google, 2024. [Online]. Available: https://ai.google.dev/edge/mediapipe/solutions/vision/hand_landmarker. Accessed: Apr. 30, 2026.
- [3] TensorFlow, "TensorFlow Lite," Google, 2024. [Online]. Available: <https://www.tensorflow.org/lite>. Accessed: Apr. 30, 2026.
- [4] Flutter, "Flutter Documentation," Google, 2024. [Online]. Available: <https://docs.flutter.dev>. Accessed: Apr. 30, 2026.

- [5] Firebase, "Firebase Documentation," Google, 2024. [Online]. Available: <https://firebase.google.com/docs>. Accessed: Apr. 30, 2026.
- [6] Apple Developer Documentation, "AVCaptureSession," Apple Inc., 2024. [Online]. Available: <https://developer.apple.com/documentation/avfoundation/avcapturesession>. Accessed: Apr. 30, 2026.
- [7] Apple Developer Documentation, "AVCaptureVideoPreviewLayer," Apple Inc., 2024. [Online]. Available: <https://developer.apple.com/documentation/avfoundation/avcapturevideopreviewlayer>. Accessed: Apr. 30, 2026.
- [8] OpenRouter, "OpenRouter Documentation," OpenRouter, 2026. [Online]. Available: <https://openrouter.ai/docs>. Accessed: Apr. 30, 2026.
- [9] Google AI for Developers, "Gemini API Documentation," Google, 2026. [Online]. Available: <https://ai.google.dev/gemini-api/docs>. Accessed: Apr. 30, 2026.
- [10] R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed. London, U.K.: Springer, 2022.